Course Name: Geographic Information System

Week 03 - Assignment 03 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10 **Total marks: 10x1= 10**

QUESTION 1

Elevation map is an example of _____ data.

a) Discrete
b) Vector
c) Continuous
d) Categorical

Answer: (c)

Detailed Solution:

An elevation map is considered continuous data because elevation changes gradually across the Earth's surface rather than existing as separate isolated values.

QUESTION 2

_____ describes the world as a space filled with discrete, identifiable units.

a) Objects
b) Fields
c) Polygons
d) Regular points

Answer: (a)

Detailed Solution:

The object model describes the world as being made up of discrete and identifiable entities such as roads, buildings, rivers, and parcels of land.

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QUESTION 3

In a vector model, _____ are usually called as nodes.

a) Lines
b) End points
c) Intermediate points
d) All of the above

Answer: (b)

Detailed Solution:

In a vector model, end points are called nodes because they define the start and end locations of line features and help establish spatial connectivity.

QUESTION 4

The degree of agreement between several measurements of a quantity is denoted as_____.

a) Accuracy
b) Discrete
c) Fuzzy
d) Precision

Answer: (d)

Detailed Solution:

Precision refers to the degree of agreement among repeated measurements of the same quantity. Higher precision means the measurements are very close to each other.

QUESTION 5

Redundancy of data can be observed in _____ model.

a) Topological
b) Spaghetti
c) Raster
d) Quad-tree

Answer: (b)

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Detailed Solution:

Data redundancy is common in the spaghetti model because spatial relationships are not stored efficiently, causing repeated storage of boundaries and coordinates.

QUESTION 6

The disadvantage of quad tree model is/are _____.

a) Processing time
b) Huge storage capacity
c) Prolong alterations
d) All of the above

Answer: (d)

Detailed Solution:

The quad-tree model may suffer from disadvantages such as increased processing time, large storage requirements, and complex alterations, therefore the correct answer is All of the above.

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QUESTION 7

_____ describe the terrain surface including the above ground objects numerically in the form of x, y, and z coordinates.

a) Digital terrain model
b) Digital surface model
c) Digital elevation model
d) All of the above

Answer: (b)

Detailed Solution:

A Digital Surface Model (DSM) represents terrain surfaces along with above-ground features such as buildings, vegetation, and other objects using x, y, and z coordinates.

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QUESTION 8

Connectivity, area definition and contiguity are basic principles of _____.

A) Topology
B) Field model
C) Raster model
D) Discrete model

Answer: (a)

Detailed Solution:

Connectivity, area definition, and contiguity are the main principles of topology, which defines spatial relationships between geographic features in GIS.

QUESTION 9

An imaginary line on the ground joining the points having the same elevation at a specific interval is known as _____.

a) Triangulated irregular network
b) Ridge line
c) Isoline
d) Valley line

Answer: (c)

Detailed Solution:

An imaginary line joining points of equal elevation is called an isoline (or contour line). It is commonly used in topographic maps to represent terrain elevation.

QUESTION 10

A lake with its exact boundary can be marked using _____ model.

a) Raster
b) TIN
c) Vector
d) DEM

Answer: (c)

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Detailed Solution:

A lake with an exact boundary is best represented using the vector model because vector polygons can accurately define precise boundaries and shapes of geographic features.

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